Exercise 4

INNER JOIN

① SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary, p.ProjectID, p.ProjectName, p.Budget, p.Status

FROM employee e

INNER JOIN projects p

ON e.EmployeeID = p.EmployeeID;

EmployeeID	FirstName	LastName	Department	Salary	ProjectName	Budget	Status
1	John	Doe	IT	80 000	Website	90 000	Ongoing
2	Sara	Smith	HR	60 000	Training	50 000	Completed

② LEFT JOIN

- SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary, p.ProjectID, p.ProjectName, p.Budget, p.Status

FROM employees e

LEFT JOIN projects p

ON e.EmployeeID = p.EmployeeID;

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	80 000	101	Website	90 000	Ongoing
2	Sara	Smith	HR	60 000	102	Training	50 000	Completed
3	Mark	Lee	IT	75 000	NULL	NULL	NULL	NULL

3 Right Join

SELECT p.ProjectID, p.ProjectName, p.Budget, p.Status, e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary

FROM employees e

RIGHT JOIN projects p

ON e.EmployeeID = p.EmployeeID;

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	Website	90 000	Ongoing	1	John	Doe	IT	80 000
103	App Dev	120 000	ongoing	NULL	NULL	NULL	NULL	NULL

4 FULL OUTER JOIN

```
SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department,
       e.Salary, p.ProjectID, p.ProjectName, p.ProjectID, p.ProjectName,
       p.Budget, p.Status
FROM employees e
FULL OUTER JOIN projects p
ON e.EmployeeID = p.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	80 000	101	Website	50 000	Ongoing
3	Mark	Lee	IT	75 000	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL	103	App Dev	120 000	Ongoing

5 UNION (Unique)

```
SELECT City AS location
FROM employees
UNION
SELECT Status
FROM projects;
```

Location
New York
Chicago
Ongoing
Completed

6. UNION ALL (Duplicates allowed)

```
SELECT City AS ~~local~~ location
FROM employees
UNION ALL
SELECT Status
FROM projects;
```

Location
New York
Chicago
New York
Ongoing
Completed

7. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM employees
WHERE Salary > 70 000;

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	80 000
3	Mark	Lee	IT	75 000

7. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM employees
WHERE Salary > 70 000;

FirstName	LastName	Department	Salary
John	Dee	IT	80 000
Mark	Lee	IT	75000

8. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM employees
WHERE Department IN ('IT', 'Finance');

Employee ID	FirstName	LastName	Department	Salary
1	John	Dee	IT	80 000
3	Mark	Lee	IT	75 000

9. SELECT ProjectID, ProjectName, Budget,
STATUS
FROM projects
WHERE Status != 'Completed';

ProjectID	ProjectName	Budget	Status
101	Website	90 000	Ongoing

10 SELECT projectID, ProjectName, Budget, Status
FROM projects
WHERE Budget > 70 000 AND Status != 'Completed';

ProjectID	ProjectName	Budget	Status
101	Website	90 000	Ongoing

11 SELECT EmployeeID, FirstName, LastName, Department, Salary, City
FROM employees
WHERE City IN ('New York', 'Toronto')
ORDER BY Salary DESC;

EmployeeID	FirstName	LastName	Department	Salary	City
1	John	Dee	IT	80 000	New York

12. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3;

EmployeeID	FirstName	LastName	Department	Salary
1	Jim	Doe	IT	80 000
3	Nick	Lee	IT	75 000

13. SELECT Department,
SUM (Salary) AS TotalSalary
FROM employees
GROUP BY Department
ORDER BY TotalSalary DESC;

Department	Total Salary
IT	155 000
HR	60 000

14. SELECT City,
AVG (Salary) AS AverageSalary
FROM employees
GROUP BY City
HAVING AVG (Salary) > 65 000;

City	Average Salary
New York	80 000

15 SELECT Department,
COUNT (EmployeeID) AS Employee Count
FROM employees
GROUP BY Department
HAVING Count (EmployeeID) > 1;

Department	Employee Count
IT	2

16. SELECT Status,
COUNT (Project ID) AS ProjectCount
GROUP BY Status
HAVING COUNT (Project ID) > 2;

Status	Project Count
Ongoing	2

17. SELECT e.EmployeeID, e.FirstName, e.LastName,
SUM (p.Budget) AS TotalProjectBudget
FROM employees e
JOIN projects p ON e.EmployeeID = p.EmployeeID
GROUP BY e.EmployeeID, e.FirstName, e.LastName
HAVING SUM (p.Budget) > 150 000;

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	160 000